COMP1521 – W8 File Systems

Notices

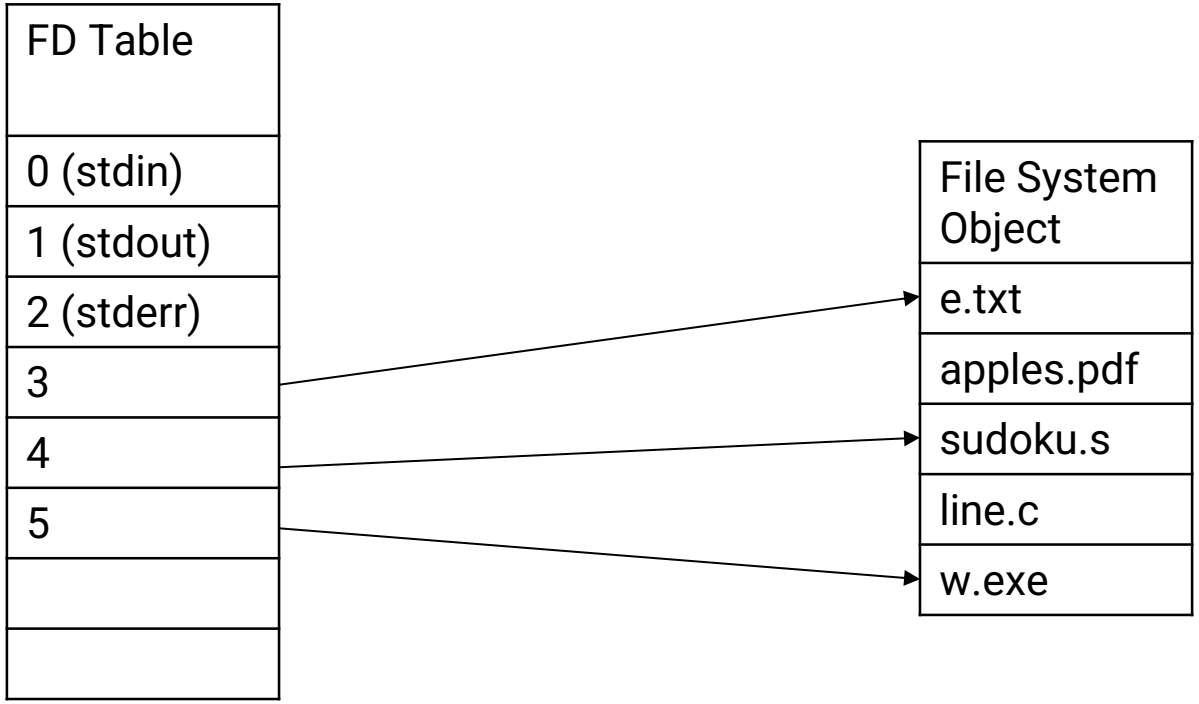
- **ASSIGNMENT 2**
  - **Due Friday W10**
- **Quiz DUE TOMORROW!**

File System Calls (UNIX)

Id	Name	Function
0	read	Read some bytes from a file **descriptor**
1	write	Write some bytes from a file **descriptor**
2	open	Open a file system object, assign it a file **descriptor**
3	close	Close a file **descriptor**
4	stat	Get file system metadata (File size, name, permissions etc)
8	lseek	Move file **descriptor** to a specified offset within a file

File Descriptors

- A file descriptor (FD) is an id for a file system object
- Anytime a file is opened, the lowest available FD is assigned to it
- A FD becomes unassigned after its corresponding file is closed

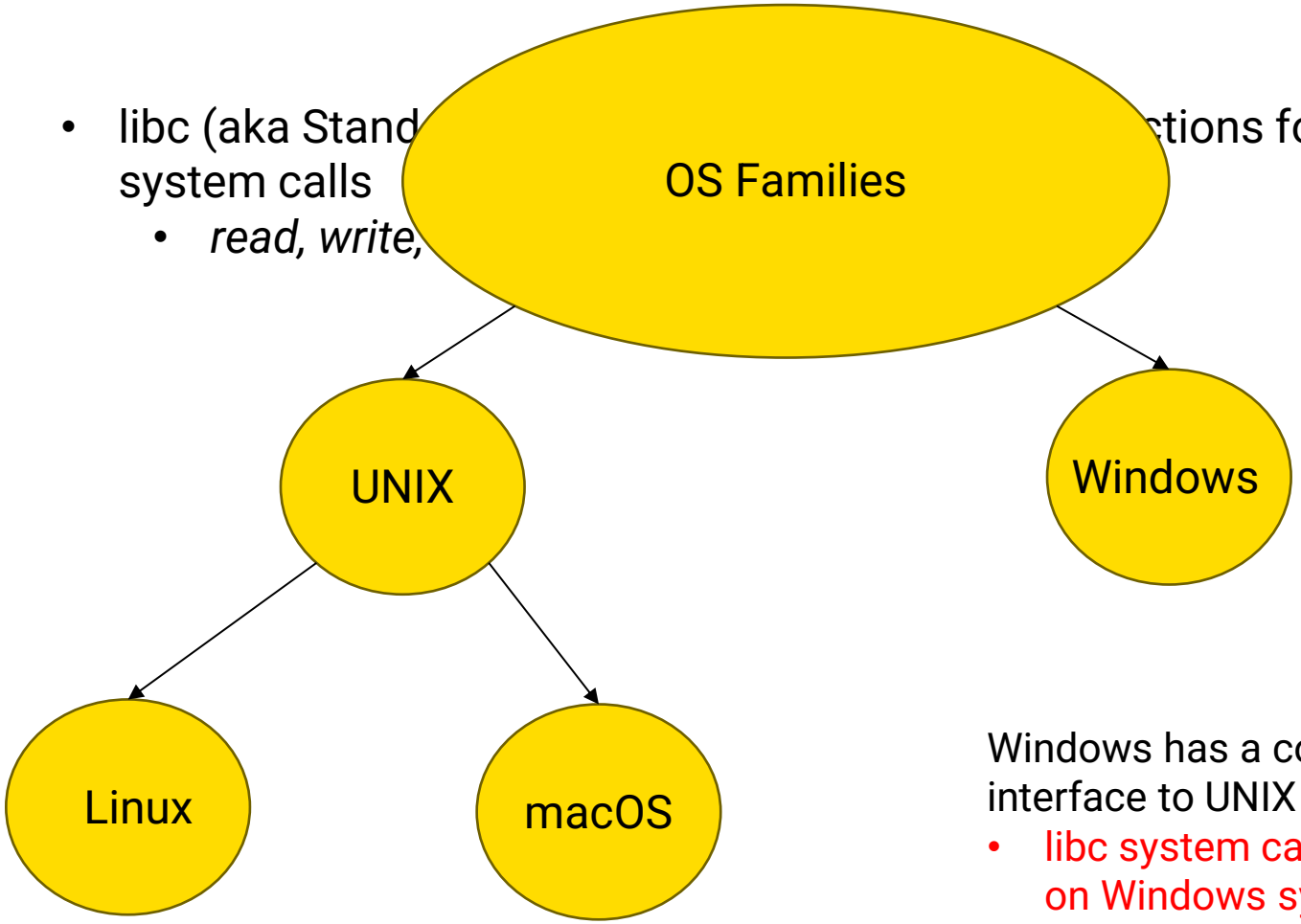


- A FD contains:
  - The current position in a file
  - File status (read, write etc)

- Note: *Multiple FDs can point to the same file*

System Calls In C

- libc (aka Stand ctions for most UNIX system calls
  - *read, write,*



Windows has a completely different interface to UNIX

- libc system call functions cannot be used on Windows systems

Solution: USE stdio functions

Standard I/O Library

- Functions are portable across operating systems
- Provides additional functions that make file manipulation easier
  - *printf(), fgetc(), fgets()* etc.

libc	stdio
open()	*fopen()*
close()	*fclose()*
read()	*fread()*
write()	*fwrite()*
lseek()	*fseek()*

Sidetrack: errno (Assignment 2 Hint)

- *How to handle syscalls?*

1. If a syscall caused the error, use *perror()* to print error message.
2. If a user-defined error has occurred, use fprintf(stderr) to print error result
3. If the error is non-fatal, continue execution. Otherwise clean up memory and *exit(1)*

errno: is a hidden global integer that indicates what went wrong

perror() or strerror() can be used to convert errno into a standard error message

Q2-Q4

Q5-Q7

Q8-Q10

UNIX Directories

UNIX Ideology: Everything is a file (or a connection between files)

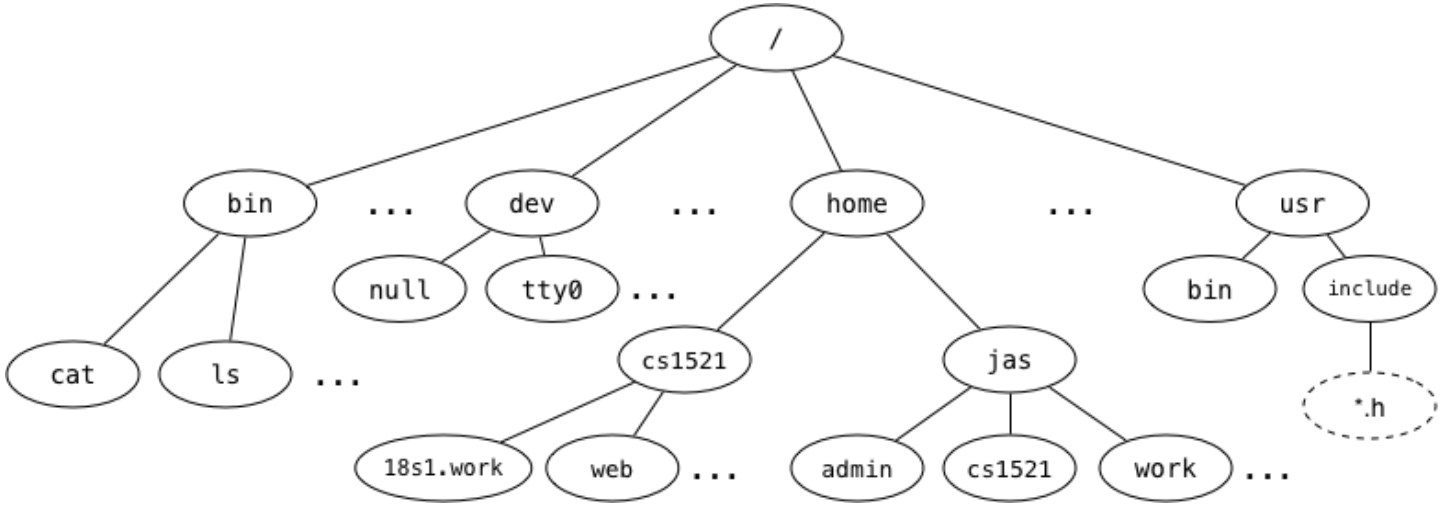
In UNIX every terminal command, every driver, every application, every internet connection should reduce to a file

Common commands like *ls, mkdir, cd* are calling files that execute said operations

UNIX Directories contain:

- files
- other directories
- symbolic links
- etc

Q1



a) What's the absolute path for the web directory? /home/cs1521/web
b) What is the link to a parent directory stored as? ..
c) Which directory is ~*jas*/../.. / - the root directory
d) What kind of object is cat? a file
e) What kind of object is home? a directory
f) What is a symbolic link? a link to a file system object

UNIX Metadata

- Each file in UNIX has metadata stored in a data structure called an inode
- Can be retrieved using the *stat()* system call

```
struct stat {
    ino_t st_ino;         /* inode number */
    mode_t st_mode;       /* protection */
    uid_t st_uid;         /* user ID of owner */
    gid_t st_gid;         /* group ID of owner */
    off_t st_size;        /* total size, in bytes */
    blksize_t st_blksize; /* blocksize for filesystem I/O */
    blkcnt_t st_blocks;   /* number of 512B blocks allocated */
    time_t st_atime;      /* time of last access */
    time_t st_mtime;      /* time of last modification */
    time_t st_ctime;      /* time of last status change */
};
```

File Permissions

- Stored as st_mode field in stat struct
- 16 bit data type

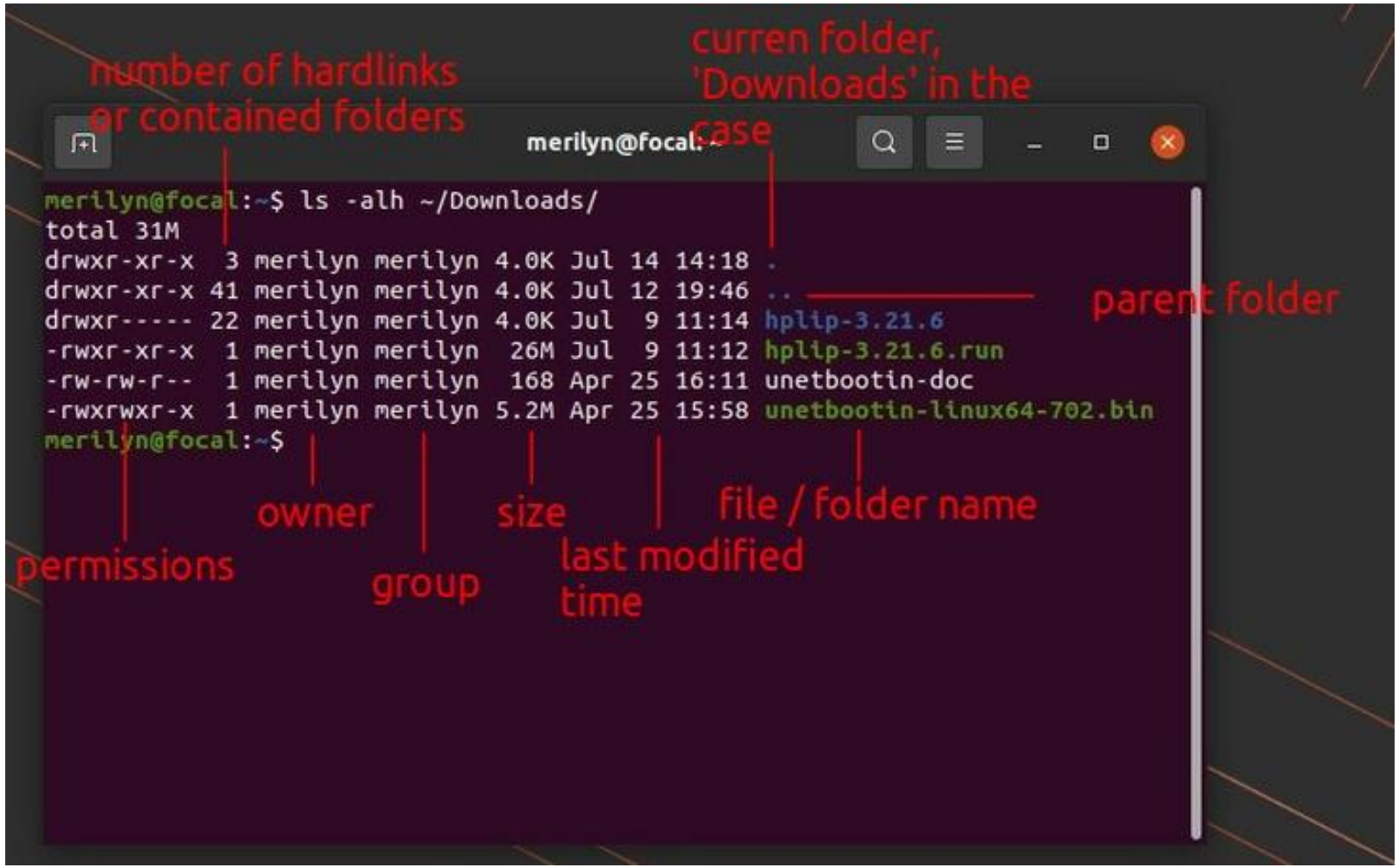
12b – 15b: File type
11b – 09b: Reserved (Out of course scope)
08b – 06b: Owner permissions
05b – 03b: Group permissions
02b – 00b: Other permissions

o100755

Object type	mask	macro	Octal
file	*S_IFREG*	*S_ISREG()*	o100000
directory	*S_IFDIR*	*S_ISDIR()*	o040000
symbolic link	*S_IFLNK*	*S_ISLNK()*	o120000
…	…		…

Permissions								
7			5			5		
1	1	1	1	0	1	1	0	1
r	w	x	r	-	x	r	-	x

Metadata


number of hardlinks
or contained folders
curren folder,
'Downloads' in the
case
merilyn@focal:~
merilyn@focal:~$ ls -alh ~/Downloads/
total 31M
drwxr-xr-x 3 merilyn merilyn 4.0K Jul 14 14:18 .
drwxr-xr-x 41 merilyn merilyn 4.0K Jul 12 19:46 ..
drwxr----- 22 merilyn merilyn 4.0K Jul 9 11:14 hplip-3.21.6
-rwxr-xr-x 1 merilyn merilyn 26M Jul 9 11:12 hplip-3.21.6.run
-rw-rw-r-- 1 merilyn merilyn 168 Apr 25 16:11 unetbootin-doc
-rwxrwxr-x 1 merilyn merilyn 5.2M Apr 25 15:58 unetbootin-linux64-702.bin
merilyn@focal:~$
parent folder
owner
size
file / folder name
permissions
group
last modified
time

Q12/13

ls variations	
ls –l	Long format
ls –li	Include inode number
ls –ln	Convert owner and group to id format
ls –lu	Provide last modified time instead of last access time